LOCAL FINITE PRESENTATION AND FLATNESS OF A FLAT SUPPORT AFTER AN ELEMENTARY ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The local structure statement for a finite type flat sheaf at a point. Same chapter flat.tex, label proposition-finite-type-flat-at-point, invoked at proof line 2865. Applied to the affine space over the base and the pushforward of the structure sheaf of the source, it gives the elementary etale neighbourhood and the open subscheme on which the pushforward is flat and of finite presentation.

Source reference. Original proposition 05I5: sources/stacks-flat/pinned-original-source.tex, source lines 2477–2477. Navigation pointer for \ref{proposition-finite-type-flat-at-point} used inside the counted span (line 2865); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be locally of finite type over S. Let $x \in X$ with image $s \in S$. If X is flat at x over S, then there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ and an open subscheme*

$$V \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$$

which contains the unique point of $X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ mapping to x such that $V \to \mathrm{Spec}(\mathcal{O}_{S',s'})$ is flat and of finite presentation.

Proof. The question is local on X and S, hence we may assume X and S are affine. Write $X = \mathrm{Spec}(B)$, $S = \mathrm{Spec}(A)$ and write $B = A[x_1, \ldots, x_n]/I$. In other words we obtain a closed immersion $i : X \to \mathbf{A}^n_S$. Denote $t = i(x) \in \mathbf{A}^n_S$. We may apply

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Proposition 05I5 to $\mathbf{A}^n_S \to S$, the sheaf $\mathcal{F} = i_*\mathcal{O}_X$ and the point t. We obtain an elementary étale neighbourhood $(S', s') \to (S, s)$ and an open subscheme

$$W \subset \mathbf{A}^n_{\mathcal{O}_{S',s'}}$$

such that the pullback of $i_*\mathcal{O}_X$ is flat and of finite presentation. This means that $V := W \cap \big(X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})\big)$ is the desired open subscheme. $\square$